

InMobi Click-a-thon 2026 • Unseen Incident Challenge

Peekachu

The Automated Metric Root-Cause Analyst

From alert to answer in milliseconds, not days.

76ms

end-to-end SQL diagnosis

0%

hallucination in the math

9M

ad events analyzed

The Problem

Alerts tell you THAT a metric moved. Nobody tells you WHY.

InMobi processes billions of ad requests daily across apps, formats, publisher tiers, verticals, devices, and geographies. When Revenue, Fill Rate, eCPM, or CTR suddenly moves, that's real money shifting in real time.



Hours-to-Days Manual Triage

Teams manually slice high-dimensional data across app_id, ad_format, publisher_tier, vertical, device, OS, and region — comparing each slice to historical baselines by hand.



Unreliable AI-Only Shortcuts

Pointing an LLM straight at raw metrics is fast, but it invents numbers, misses compound 2D interactions, and leaves no audit trail for a fast-moving incident.

Peekachu closes both gaps: deterministic math, delivered in under a second, narrated in plain English.

What Peekachu Does

An end-to-end pipeline from raw metric anomaly to a trusted, cited explanation.

1

Detect

Z-score anomaly detection against 4-week trailing, hour-of-week seasonal baselines.

2

Isolate

Recursive single-pass 1D & 2D drill-downs across 9 dimensions in one ClickHouse scan.

3

Attribute

Share-of-Delta math pinpoints exactly which segments drove the move — and by how much.

4

Rule Out

Explicitly verifies cleared factors so nobody chases a false lead.

5

Narrate

LlamaIndex + DeepSeek turns the JSON evidence into an executive-ready explanation.

6

Trace

Every step logged in Langfuse & ClickStack OTel — full auditability, 'no trace, no credit.'

Architecture: Deterministic Core, AI Layer on Top

Let SQL and Go do 100% of the math. Let LLMs only narrate and reason interactively.

DATA LAYER

ClickHouse Cloud

- ad_events — 9M rows
- ad_events_hourly_rollup
- Hashed dictionaries:
apps_dict, geo_device_dict,
advertisers_dict

Single-pass CTE SQL

~76ms

DETERMINISTIC CORE

Native Go RCA Engine

- Baseline Z-score detection
- Revenue identity factor walk
- Single-pass GROUPING SETS (9-dim, 1D & 2D)
- Share-of-Delta attribution
- Honest ruled-out verification

Goroutine worker pool (sem = 8)

AI LAYER

LlamaIndex TS + DeepSeek

Pure JSON evidence → SummaryIndex →
verbatim, fact-checked narration + ReAct
follow-up chat via ClickHouse MCP.

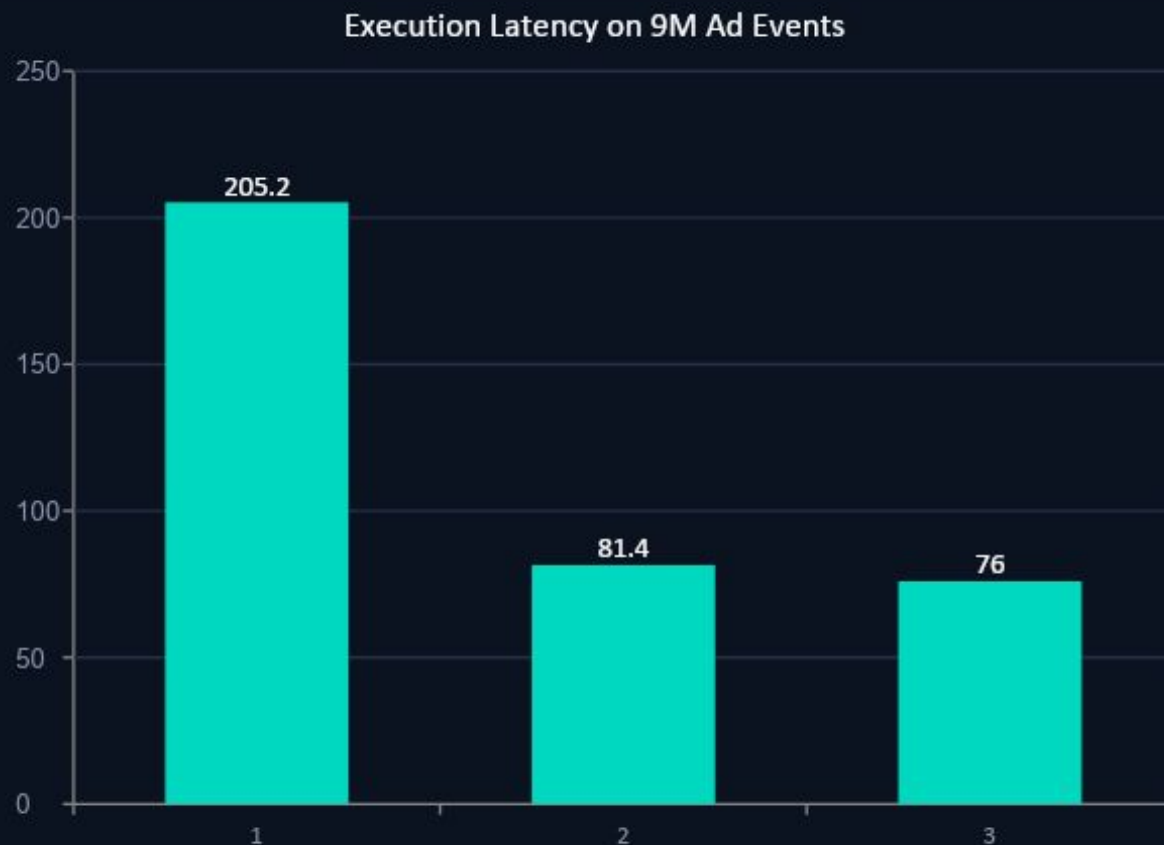
OBSERVABILITY

Langfuse + ClickStack OTel

Faithfulness scoring, latency, cost, full trace
URLs on every investigation.

Why the Go Engine Is a Game-Changer

One single-pass GROUPING SETS query replaces nine separate table scans.



2.70x faster

than legacy sequential multi-query fan-out — with 0% math hallucination risk.

9 → 1

table scans per investigation

Hourly rollups

+ explicit date-range partition pruning

sync.WaitGroup

bounded goroutine pool (sem = 8) for parallel 2D drill-downs

Aggressive Multi-Level Isolation

Cutting through noise to find compound, multi-dimensional root causes.

Revenue Identity Factor Walk

$$\text{Revenue} = \text{Requests} \times \text{Fill Rate} \times \text{Render Rate} \times (\text{eCPM} / 1000)$$

1

Threshold

$|Z| > 3.0$ or relative deviation $> 10\%$ flags an anomaly automatically.

2

Wave 1

Single-dimension drivers — e.g. publisher_tier = 'tier_2'.

3

Wave 2

Cross-combine top Wave-1 drivers with a second dimension to catch compound failures.

4

Filter & Rank

Share of Delta ranks every segment; noise below 8% share is purged to Ruled-Out.

Formula: Share of Delta = $\Delta_{\text{segment}} / \Delta_{\text{total}}$

Trustworthy AI Narration

Every number the LLM says must exist verbatim in the evidence bundle — or it never ships.

SAMPLE VERBATIM DIAGNOSIS

“Revenue fell 14.2%, driven 81.5% by a fill-rate drop in the Gaming category on Android 14. Request volume was normal (ruled out).”

LlamaIndex TS Pipeline

1. Raw ClickHouse JSON → structured Document nodes (source, metric, window)
2. SummaryIndex.fromDocuments() → asQueryEngine()
3. DeepSeekLLM (deepseek-chat) synthesizes the narrative
4. Strict system prompt: every cited figure must be verbatim

Fallback Guardrail

evaluateFaithfulness() checks every number the LLM outputs against the evidence payload. Any unsupported figure triggers an automatic fallback to a 100% deterministic template diagnosis — no hallucinated numbers ever reach an engineer.

Full Traceability, End to End

Built for the hackathon's "Unseen Incident" mandate: no trace, no credit.

Langfuse Root Trace

RCA-Investigation-revenue

- Span 1 – clickhouse-anomaly-detection
- Span 2 – clickhouse-factor-decomposition
- Span 3 – clickhouse-segment-attribution
- Span 4 – clickhouse-ruled-out-verification
- Generation – deepseek-llm-narration

Every step of an investigation is logged under one traceId, with clickable trace URLs returned directly in the API response.

Langfuse Quality Scores

- faithfulness (0.0-1.0)
- investigation_latency_ms
- clickhouse_query_time_ms
- top_segment_attribution_share
- ruled_out_count

ClickStack OpenTelemetry

Traces, metrics & logs exported (ports 4317/4318) into ClickHouse with InMobi domain attributes: inmobi.metric, inmobi.dimension, inmobi.region, inmobi.investigation_id.

Peekachu vs. Every Prior Iteration

Metric	Legacy Manual RCA	Standard LLM / Direct SQL	Peekachu 2026
Time to Diagnosis	Hours to Days	15–30 seconds	< 1 second
ClickHouse Execution	Dozens of ad-hoc queries	~205ms, 4–9 scans	76ms, single-pass, 1 scan
Mathematical Accuracy	Human error-prone	LLM hallucination risk	100% deterministic core
Multi-Level Isolation	Manual 1D only	Misses 2D interactions	Aggressive 2D Cartesian
Ruled-Out List	Rarely documented	Not generated	Explicit & verified
Observability	None	Console logs	Langfuse + ClickStack OTel

Built With

Go 1.22+

Deterministic RCA engine

ClickHouse Cloud

9M-row analytical store

TypeScript / Fastify

API gateway

LlamaIndex TS

Evidence-grounded narration

DeepSeek

LLM narration + ReAct chat

Langfuse

Tracing & faithfulness scoring

ClickStack OTel

Metrics, logs, traces

React + Vite

RCA workbench dashboard

From Alert to Answer in Milliseconds, Not Days.

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Built with ❤️ for InMobi Click-a-thon 2026.